

tea1 の話

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2026-04-08

スライドはこちら

- https://nissinbo.github.io/RwPL_1_tea/



自己紹介



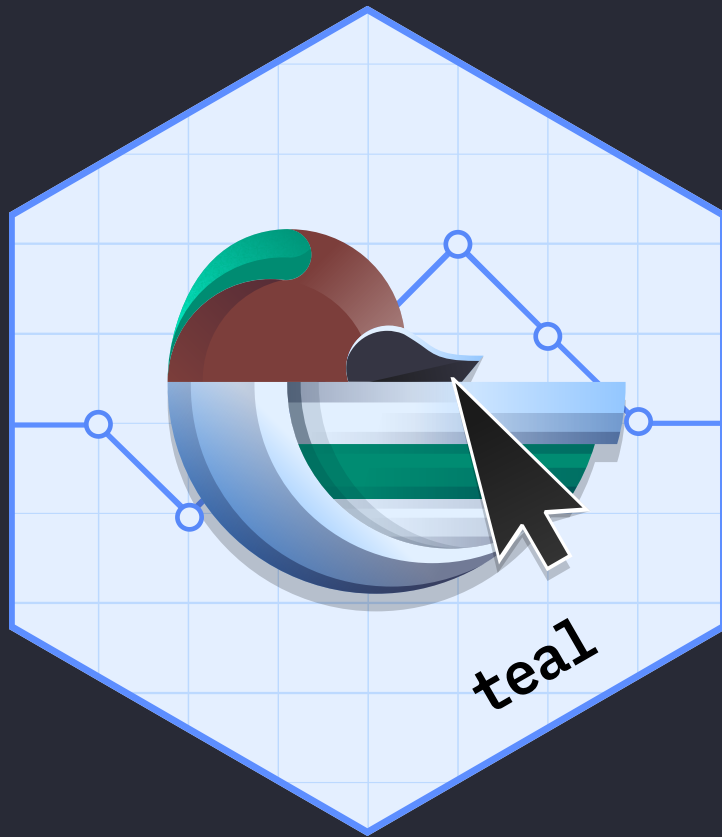
- 西川寛来
 - 製薬業界で修行中
 - LinkedIn
 - GitHub

こんな話をします 🙋

- teal とはなにか
- “あの日の CRAN” を使う
- OSS contribution について

teal とはなにか

teal



- Shiny で治験データ分析ダッシュボードを作るための R パッケージ
 - GitHub
 - Roche/Genentech 社の **Insights Engineering** が開発
- teal を用いた FDA submission の検証が行われた
 - Submission Pilot 2, Shiny app
 - Submission Pilot 4, Shiny app

活用場面

- データモニタリング
- 照会事項対応
- 事後解析
- 探索的な解析
- (将来的には) 承認申請

 治験データのダッシュボードでやることは多い

製薬企業の採用事例

- GSK
 - PHUSE US Connect 2024
- Sanofi
 - PHUSE US Connect 2025
- BMS, JnJ, Sanofi, Roche
 - PHUSE Working Group Events

これ以外にも事例は増えている印象。日本でもやっていきましょう

teal で作った Shiny はこんな感じ 🙋

teal.gallery efficacy

The screenshot shows a web browser window displaying a Shiny application titled "My first teal app" by "NEST @ Roche". The browser tabs include "Simplenote", "Efficacy Analysis Teal Demo App", and "Patient Profile Analysis Teal Den". The address bar shows the URL "rinpharma.shinyapps.io/nest_efficacy_stable/".

The application interface includes a top navigation bar with "Module (15)" and "Report" dropdowns. Below this, the breadcrumb "Home / Kaplan Meier Plot" is visible. On the right, there are buttons for "+ Add to Report" and "Show R code".

The main content area is divided into several panels:

- Active Data Summary:** A table showing data for ADSL and ADTTE datasets.
- Encodings:** A section for selecting datasets and endpoints.
- Analysis Variable:** A section for selecting the analysis variable (AVAL).
- Censor Variable:** A section for selecting the censor variable (CNSR).
- Facet Plots by Dataset:** A section for selecting facet plots by dataset (ADSL).
- AGEGR1 Age Group:** A section for selecting the age group (AGEGR1).
- Select Treatment Variable:** A section for selecting the treatment variable.
- ARM Description of FX:** A section for selecting the arm description.
- Compare Treatments:** A toggle switch for comparing treatments.

A large central plot area displays a message: "Some inputs require attention". Below this message, three specific requirements are listed:

- A reference arm must be selected
- An analysis variable is required
- A censor variable is required

At the bottom left, a notification bar indicates "Disconnected from the server." with a "Reload" button and a video player progress bar showing "0:00 / 1:19".

実装の雰囲気

```
app.R
1 # load libraries
2 library(teal)
3 library(teal.modules.general)
4 library(teal.widgets)
5 library(sparkline)
6
7 # teal_data object
8 data <- teal_data()
9 data <- within(data, {
10   ADSL <- teal.data::rADSL
11   ADTTE <- teal.data::rADTTE
12 })
13 join_keys(data) <- default_cdisc_join_keys[names(data)]
14
15 # initialize the app
16 app <- init(
17   data = data,
18   modules = modules(
```

- 治験データで行われる分析手法をモジュールとして利用可能
 - 裏では **tern** パッケージが動いている

teal を使うメリット

インタラクティブな探索

- ユーザー自身でフィルタの操作や、カスタムレポートを出力
- 解析部門の負担を減らし、エンドユーザーの満足度を高める

コード管理による効率性と再現性

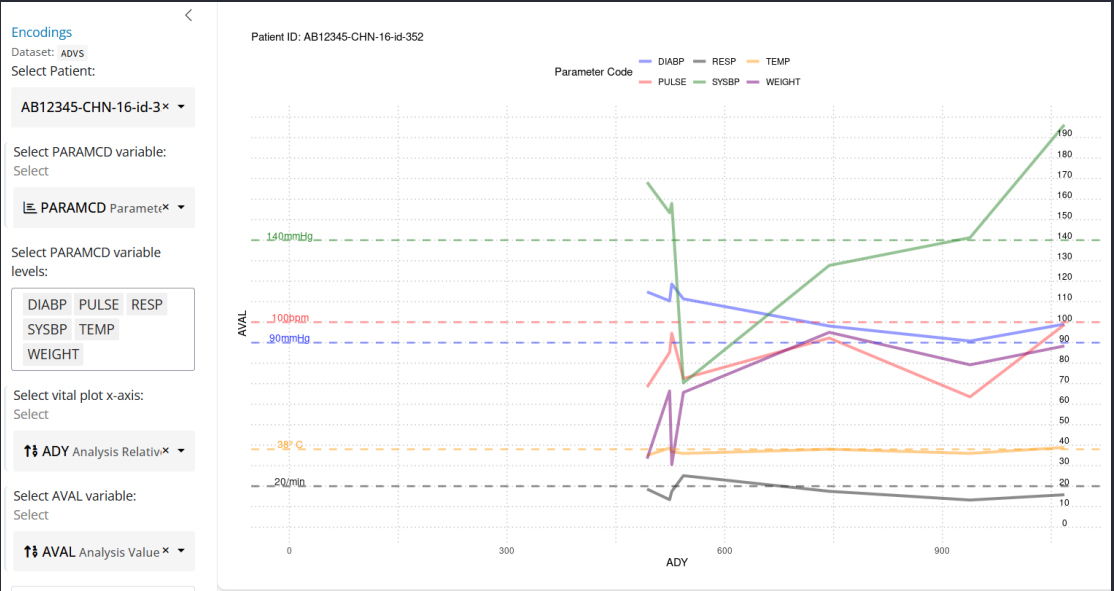
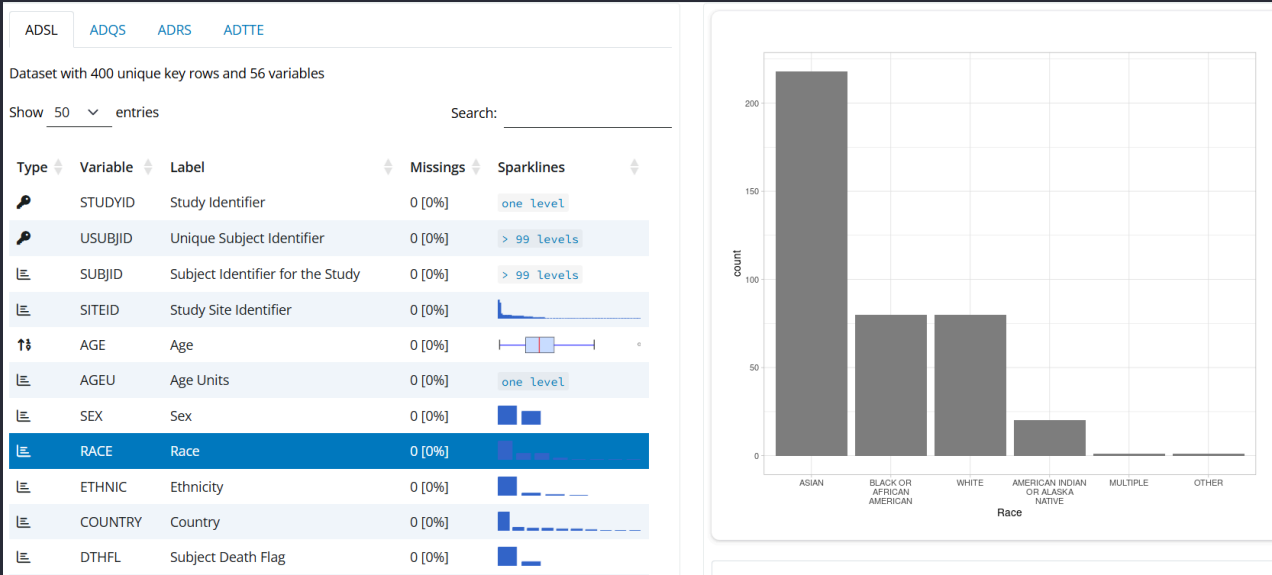
- 試験間での横展開がしやすい
- アプリ上の操作を再現する R コードを出力

モジュールによる標準化

- 生物統計に特化しており、要求に素早く対応
- ダッシュボードの作り方を揃えやすい

モジュールの探し方

- teal.modules.* のパッケージで各モジュールが利用可能
 - teal.modules.general, teal.modules.clinical



Encodings

Datasets: ADS, ADSL

Parameter Dataset: ADS

Filter by

BESRSPI Best Confirme

Responders

CR PR

Select Treatment Variable

Dataset: ADSL

ARM Description of F

Compare Treatments

Groups

Ref: A: Drug X

Comp: B: Placebo

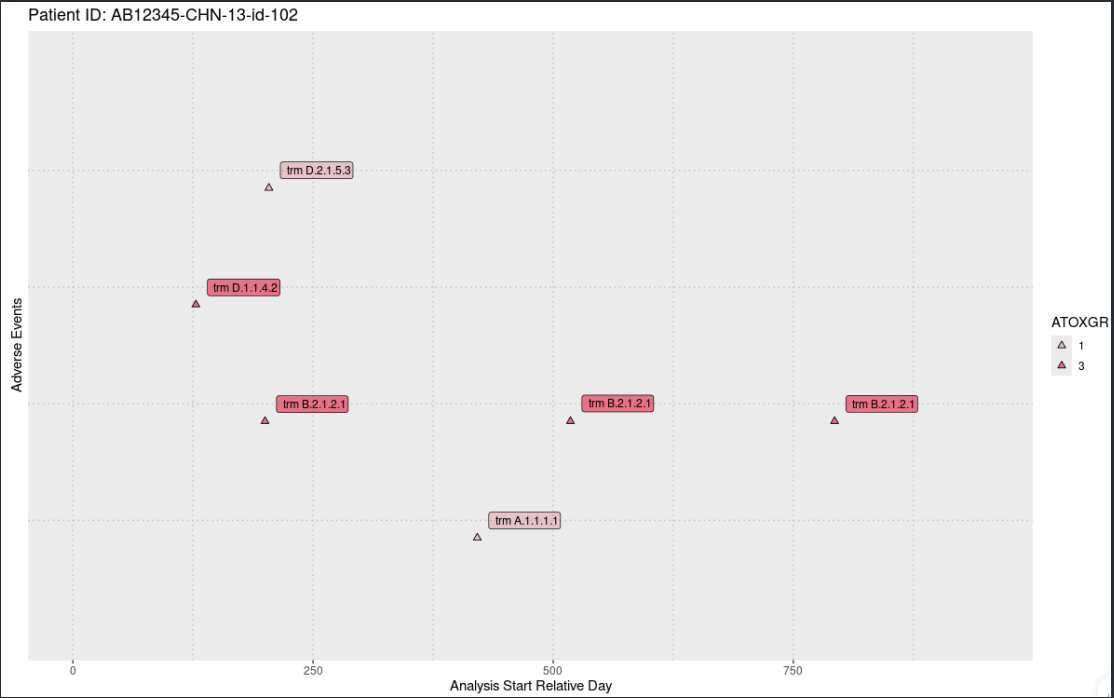
C: Combination

Multiple reference groups are automatically combined into a single group.

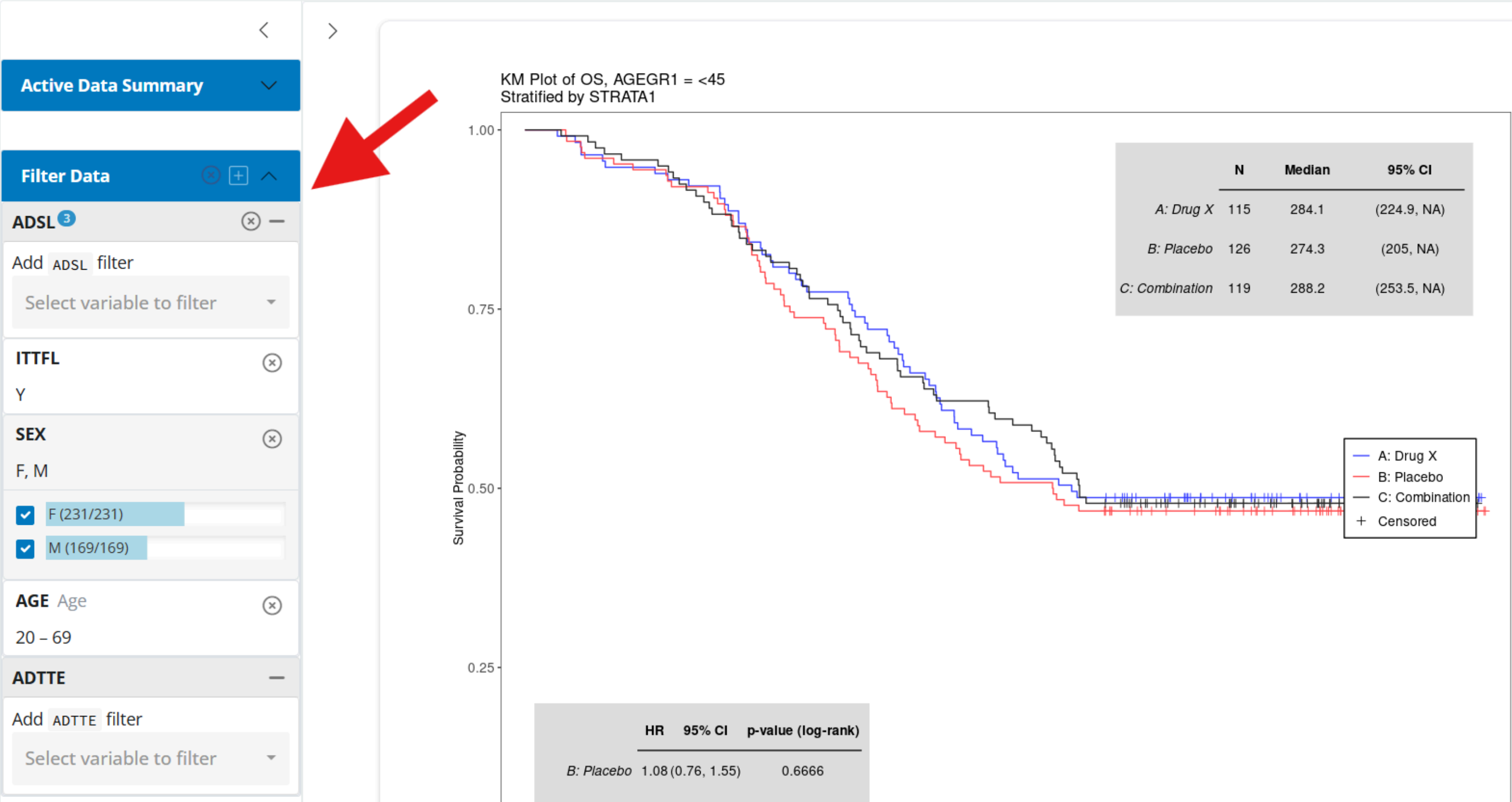
Table of BESRSPI for CR and PR Responders

Stratified by STRATA1

	A: Drug X (N=133)	B: Placebo (N=141)	C: Combination (N=126)
Responders	131 (98.5%)	134 (95.0%)	126 (100.0%)
95% CI (Clopper-Pearson)	(94.7, 99.8)	(90.0, 98.0)	(97.1, 100.0)
Unstratified Analysis			
Difference in Response rate (%)		-3.5	1.5
95% CI (Wald, without correction)		(-7.6, 0.7)	(-0.6, 3.6)
p-value (Chi-Squared Test with Schouten Correction)		0.1501	0.3010
Odds Ratio (95% CI)		0.29 (0.06 - 1.43)	35463634.84 (0.00 - Inf)
Stratified Analysis			
Difference in Response rate (%)		-3.5	1.5
95% CI (CMH, without correction)		(-7.5, 0.6)	(-0.6, 3.6)
p-value (Cochran-Mantel-Haenszel Test)		0.1097	0.1625
Odds Ratio (95% CI)		0.29 (0.06 - 1.44)	6149252.81 (0.00 - Inf)
CR			
95% CI (Clopper-Pearson)	115 (86.5%) (79.46, 91.78)	112 (79.4%) (71.82, 85.77)	116 (92.1%) (85.89, 96.13)



UI で簡単にデータをフィルタリング



カスタムレポートを作る

Module (15) Report

Home / Time To Event Table

Click here to add this module's output to the report.

[+ Add to Report](#) [Show R code](#)

Active Data Summary

Data Name	Obs	Subjects
ADSL	400/400	400/400
ADTTE	2000/2000	400/400

Filter Data

ADSL 3 ITTFL Y SEX F, M AGE Age 20 - 69 ADTTE

Encodings

Datasets: ADSL, ADTTE

Select Endpoint Dataset: ADTTE

Filter by

OS Overall Survival x

Analysis Variable

Select AVAL

Censor Variable

Select CNSR

Select Treatment Variable

Dataset: ADSL

Select

ARM Description of P_x

☐ Compare Treatments

☐ Add All Patients column

Time Points

182

Event Description Variable

Time-To-Event Table for OS

	A: Drug X (N=133)	B: Placebo (N=141)	C: Combination (N=126)
Patients with event (%)	70 (52.6%)	71 (50.4%)	67 (53.2%)
Earliest contributing event			
Death	70	71	67
Patients without event (%)	63 (47.4%)	70 (49.6%)	59 (46.8%)
Time to Event (DAYS)			
Median	282.5	287.8	287.3
95% CI	(224.9, <Missing>)	(225.9, <Missing>)	(251.2, <Missing>)
25% and 75%-ile	170.0, <Missing>	139.9, <Missing>	157.4, <Missing>
Range	16.9 to 497.0 {1}	18.2 to 499.0 {1}	18.8 to 495.0 {1}
182 DAYS			
Patients remaining at risk	95	95	86
Event Free Rate (%)	71.43	67.38	68.25

表やグラフを組み合わせてレポートを出力可能 (html, pdf, pptx, docx)

アプリ上の操作をコードとして出力

My first teal app

Module (15) Report

Home / Binary Response

NEST @ Roche

Active Data Summary

Data Name	Obs	Subjects
ADSL	400/400	400/400
ADRS	800/800	400/400

Filter Data

ADSL 3 ITTFL Y

Encodings

Datasets: ADRS, ADSL
Parameter Dataset: ADRS
Filter by
BESRSPI Best Confirmer^x

Responders

CR PR

Select Treatment Variable
Dataset: ADSL
Select
ARM Description of FX
Compare Treatments

Table of BESRSPI for CR and PR Responders
Stratified by STRATA1

	A: Drug X (N=133)	B: Placebo (N=141)	C: Combination (N=126)
Responders	131 (98.5%)	134 (95.0%)	126 (100.0%)
95% CI (Wald, with correction)	(96.1, 100.0)	(91.1, 99.0)	(99.6, 100.0)

Handwritten red annotations: A bracket and an arrow pointing to the "Show R code" button.

Buttons: Add to Report, Show R code

https://rinpharma.shinyapps.io/nest_efficacy_stable/

コードをコピペして実行すると...

My first teal app

Module (15) Report

Home / Binary Response

Active Data Summary

Data Name	Obs	Subjects
ADSL	400/400	400/400
ADRS	800/800	400/400

Filter Data

ADSL

ITTFL

Y

SEX

F, M

AGE Age

20 ~ 69

ADRS

Show R Code

Dismiss Copy to Clipboard

```

library(dplyr)
library(random.cdisc.data)
library(nestcolor)
library(sparkline)
ADSL <- radsl(seed = 1)
.adsl_labels <- teal.data::col_labels(ADSL, fill = FALSE)
.char_vars_asl <- names(Filter(isTRUE, sapply(ADSL, is.character)))
.adsl_labels <- c(.adsl_labels, AGEGR1 = "Age Group")
ADSL <- ADSL %>% mutate(AGEGR1 = factor(case_when(AGE < 45 ~ "<45", AGE >= 45 ~ ">=45"))) %>% mutate_
teal.data::col_labels(ADSL) <- .adsl_labels
ADRS <- radrs(ADSL, seed = 1)
.adrs_labels <- teal.data::col_labels(ADRS, fill = FALSE)
ADRS <- filter(ADRS, PARAMCD == "BESRSPI" | AVISIT == "FOLLOW UP")
teal.data::col_labels(ADRS) <- .adrs_labels
stopifnot(rlang::hash(ADSL) == "a38e9170aea81f1199dfe480c202142e") # @linksto ADSL
stopifnot(rlang::hash(ADRS) == "b0d502767c7e9f7b2d9118c30570b61a") # @linksto ADRS
.raw_data <- list2env(list(ADSL = ADSL, ADRS = ADRS))
lockEnvironment(.raw_data) # @linksto .raw_data
ADRS <- dplyr::inner_join(x = ADRS, y = ADSL[, c("STUDYID", "USUBJID"), drop = FALSE], by = c("STUDYID", "USUBJID"))
library(magrittr)
ANL_1 <- ADSL %>% dplyr::select(STUDYID, USUBJID, ARM, STRATA1)
ANL_2 <- ADRS %>% dplyr::filter(PARAMCD == "BESRSPI") %>% dplyr::select(STUDYID, USUBJID, PARAMCD, AVISIT)
ANL_3 <- ADRS %>% dplyr::select(STUDYID, USUBJID, PARAMCD, AVISIT, AVALC)
ANL <- ANL_1
ANL <- dplyr::inner_join(ANL, ANL_2, by = c("STUDYID", "USUBJID"))
ANL <- dplyr::inner_join(ANL, ANL_3, by = c("STUDYID", "USUBJID", "PARAMCD", "AVISIT"))
ANL <- ANL %>% teal.data::col_relabel(ARM = "Description of Planned Arm", PARAMCD = "Parameter Code",
library(magrittr)
ANL_ADSL_1 <- ADSL %>% dplyr::select(STUDYID, USUBJID, ARM, STRATA1)
ANL_ADSL <- ANL_ADSL_1
ANL_ADSL <- ANL_ADSL %>% teal.data::col_relabel(ARM = "Description of Planned Arm", STRATA1 = "Strati
anl <- ANL %>% dplyr::mutate(ARM = droplevels(ARM)) %>% dplyr::mutate(is_rsp = dplyr::if_else(!is.na(
ANL_ADSL <- ANL_ADSL %>% dplyr::mutate(ARM = droplevels(ARM)) %>% tern::df_explicit_na(na_level = "<M
lyt <- rtables::basic_table(show_colcounts = TRUE, title = "Table of BESRSPI for CR and PR Responders
table <- rtables::build_table(lyt = lyt, df = anl, alt_counts_df = ANL_ADSL)
table

```

NEST @ Roche

+ Add to Report Show R code

Placebo	C: Combination
(N=141)	(N=126)
1 (95.0%)	126 (100.0%)
1, 99.0)	(99.6, 100.0)

結果が再現された！

► Code

Table of BESRSPI for CR and PR Responders
Stratified by STRATA1

	A: Drug X (N=133)	B: Placebo (N=141)	C: Combination (N=126)
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95% CI (Wald, with correction)	(96.1, 100.0)	(91.1, 99.0)	(99.6, 100.0)

teal の歴史を歩む

v0.16.0 -> v1.0.0 にメジャーアップデートした際の Changelog を見てみる

teal 1.0.0

CRAN release: 2025-08-20

Breaking changes

- The `reporter_previewer_module()` is deprecated and will be removed in the future release. The custom `server_args` values can be set using `options()`:
 - `teal.reporter.nav_buttons` to control which buttons will be displayed in the “Report” drop-down.
 - `teal.reporter.rmd_output` to customize the R Markdown outputs types for the report.
 - `teal.reporter.rmd_yaml_args` to customize the widget inputs in the download report modal.
 - `teal.reporter.global_knitr` to customize the global `knitr` options for the report.

Enhancement

- Improved the layout and appearance of the app using `bslib` components.
- General repositioning of key navigation components across the app:
 - Modules: The module navigation is moved from a nested tab selection to a “Module” drop-down selection. The module selection can be done from the nested button of modules.
 - Reporter: The Report previewer is no longer displayed as yet another teal module, it is placed inside a “Report” drop-down right next to the “Module” drop-down. The Report previewer is a modal that displays the added report cards. The “Report” drop-down also contains other global report options like download/load/reset the Report.
 - Teal Data Module: The UI created by the `teal_data_module()` is now placed inside a modal that can only be closed when it returns `teal_data`.

v1.0.0 で何が変わった？

「UI を大幅刷新、**bslib** 使ってレイアウト改善したよ」 とのこと

teal 1.0.0

CRAN release: 2025-08-20

Breaking changes

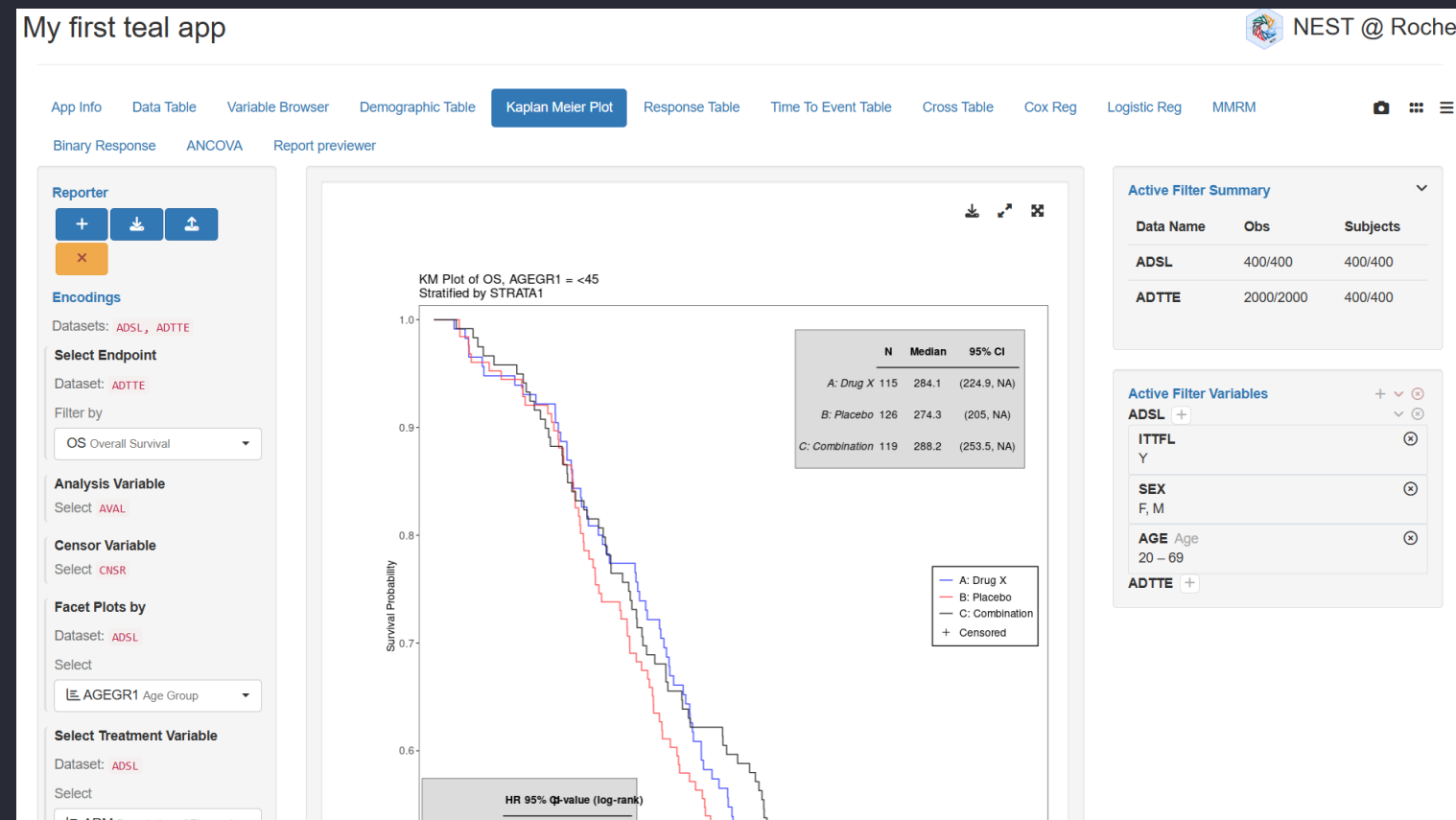
- The `reporter_previewer_module()` is deprecated and will be removed in the future release. The custom `server_args` values can be set using `options()`:
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 - `teal.reporter.rmd_yaml_args` to customize the widget inputs in the download report modal.
 - `teal.reporter.global_knitr` to customize the global `knitr` options for the report.

Enhancement

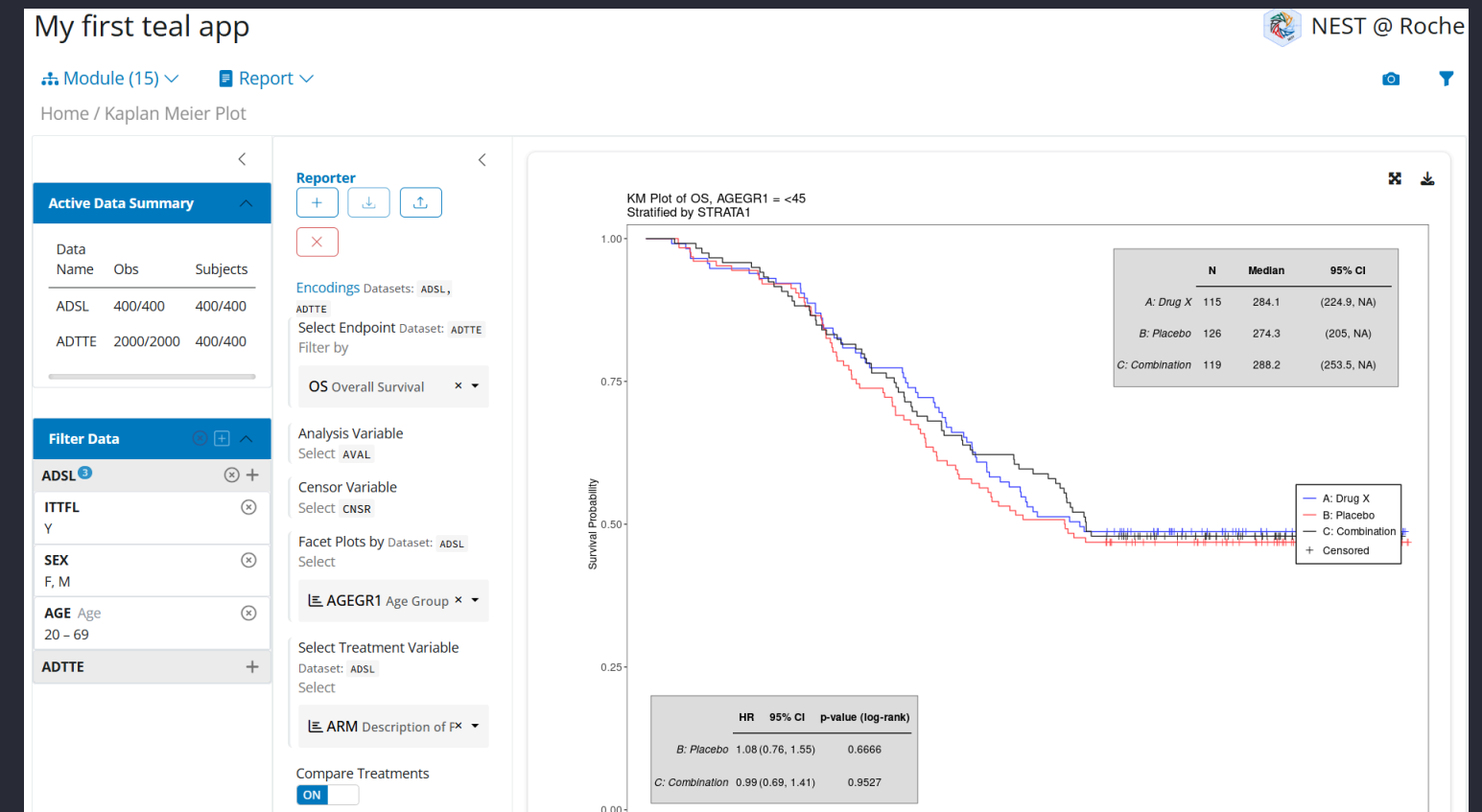
- Improved the layout and appearance of the app using **bslib** components. 🧐🧐
- General repositioning of key navigation components across the app:
 - Modules: The module navigation is moved from a nested tab selection to a “Module” drop-down selection. The module selection can be done from the nested button of modules.
 - Reporter: The Report previewer is no longer displayed as yet another teal module, it is placed inside a “Report” drop-down right next to the “Module” drop-down. The Report previewer is a modal that displays the added report cards. The “Report” drop-down also contains other global report options like download/load/reset the Report.
 - Teal Data Module: The UI created by the `teal_data_module()` is now placed inside a modal that can only be closed when it returns `teal_data`.

UI を比べてみる

v0.16.0



v1.0.0



- めっちゃ変わった!!!
 - 全体的にモダンなデザイン
 - モジュール選択がタブからドロップダウンに
 - フィルタパネルの位置が変わった

“あの日の CRAN” を使う

バージョンを指定してインストールはできるけど...

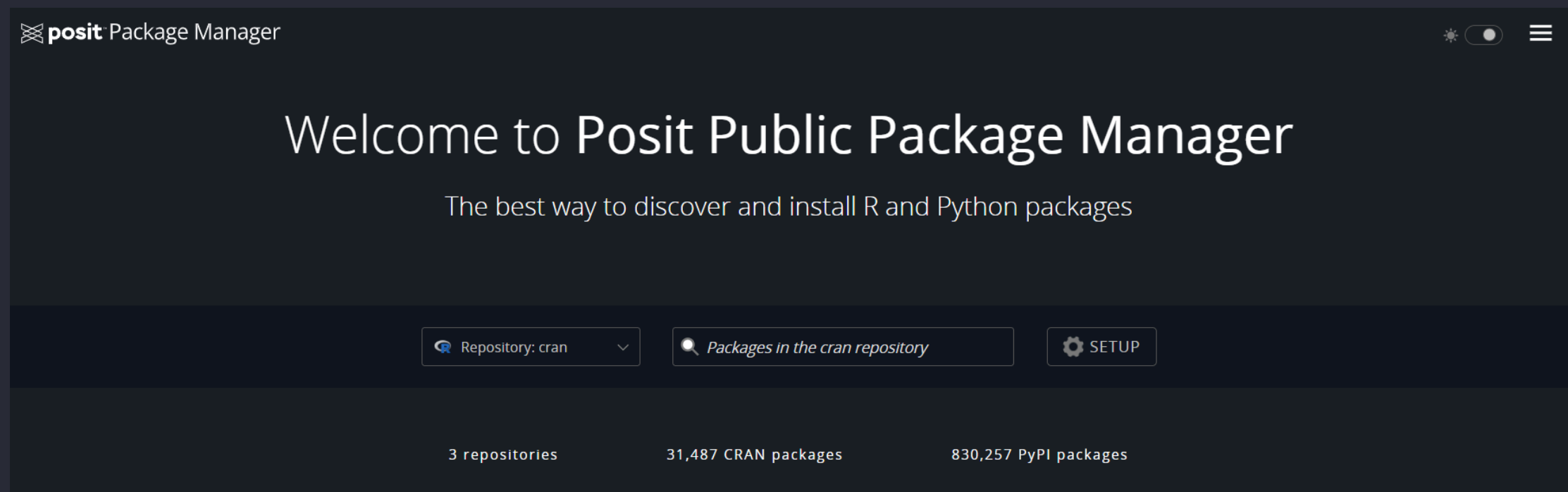
パッケージのインストールには `pak::pak()` が便利です。使いましょう。

```
1 # バージョンを指定してインストール
2 pak::pak("teal@1.0.0")
```

- デフォルトでは **CRAN binary archive** から指定バージョンをインストール
- 指定したパッケージ以外は「本日の CRAN」を参照する
 - 依存パッケージはどうなる？
 - 複数パッケージをバージョン指定してダウンロードしたい時は？

😬 依存パッケージなども含めて“当時の環境”をそのまま再現したい

P3M とは？



- Posit Public Package Manager (P3M)
 - Posit 社が提供する パッケージリポジトリ
 - 指定した日付の CRAN snapshot を利用可能
 - Windows の例: <https://packagemanager.posit.co/cran/2025-08-21>

P3M を使って“あの日の CRAN”を使う

`pak::repo_add()` や `pak::repo_resolve()` で CRAN snapshot をリポジトリとして追加する

```
1 # パッケージのバージョンを指定すると、CRAN リリースの翌日の snapshot を参照する
2 pak::repo_add(CRAN = pak::repo_resolve("PPM@teal-1.0.0"))
3
4 # 直接日付を指定してもOK
5 pak::repo_add(CRAN = "PPM@2025-08-21")
6
7 # 指定した CRAN snapshot でインストール
8 pak::pak("teal")
```

 “あの日の CRAN” を使うテクニックは、製薬業界や承認申請周りで重要になりそう

OSS contribution について

teal に貢献してみよう

- コミュニティが活発
 - プレゼンテーション・ウェビナーも頻繁に開催
 - **Pharmaverse Slack** のチャンネルもある
- 機能追加の要望が多い
- バグ報告 (= 伸びしろ) もまだまだある
- 開発チームが親切で、初心者歓迎の雰囲気
- 他社の統計プログラマーとの協働の機会

OSS contribution の流れ

1. 興味のある issue を見る
 - good first issue から選ぶとよさそう
 - 例: `ggplot2` の deprecated な関数を置き換える
2. Fork して修正・動作確認
3. Pull Request 送信
4. レビュー対応 → マージ 🎉

PR を送ってみた

- data table の重複削除で余計な列が入る問題
- AE summary table のフィルタのバグ

些細な貢献かもしれないが、自分が使っているツールを自分で改善する体験は楽しい

まとめ

やっつけていきましょう

teal

- 治験データのダッシュボード作成に最適
- 多くの企業で採用実績あり
- エンドユーザーが自分でレポートを作れる
- P3M で“あの日の CRAN” を使おう

OSS contribution

- まずは good first issue から
- 自分が使っているツールを自分で改善できる

Enjoy 🙌

紹介しきれなかった teal お役立ちリンク

- 大阪SAS勉強会: Be lazy with teal
- YouTube: teal workshop
- PHUSE US Connect 2024 Roche
- PHUSE US Connect 2025 CIM Global
- PHUSE US Connect 2025 Roche
- PHUSE US Connect 2026 Pfizer
- YouTube: Implementation of {teal} Shiny apps in DMC activity
- Appsilon Blog: Innovating pharma with teal
- Appsilon Blog: tealflow